

Salem Werdyani, Ph.D.

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Professional Summary

Distinguished scientist with a strong background in human genetics, bioinformatics, AI-based research analyses, and statistical genomics. Proficient in designing and applying machine and deep learning approaches to unravel the genetic architecture of complex diseases. Demonstrated expertise in developing predictive models from large-scale GWAS, WES, WGS, and single-cell RNA-seq data, integrating multi-omics and clinical datasets to generate actionable insights for patient stratification, drug discovery, and precision medicine. Established record of securing competitive funding and publishing over ten peer-reviewed papers. Experienced leader in multidisciplinary research environments with proven ability to mentor early-career scientists, lead cross-functional teams, and build strong collaborative networks. Skilled in driving multiinstitutional research initiatives, optimizing analytical workflows, and managing timelines to achieve strategic objectives. Adept at overseeing research budgets, coordinating large-scale projects, and effectively communicating complex scientific findings to academic, clinical, and policy audiences.

Education

Ph.D. in Medicine (Human Genetics) | Memorial University of Newfoundland (2018 – 2023)

Dissertation: Genomic and metabolomic studies for better understanding osteoarthritis pathogenesis.

M.Sc. in Medicine (Human Genetics) | Memorial University of Newfoundland (2012 – 2016)

Thesis: Characteristics of the germline Copy Number Variations (CNVs) in colorectal cancer patients.

Post-Graduate Certificate in Bioinformatics | Centennial College, Toronto, ON (2010 – 2011)

B.Sc. Biology | United Arab Emirates University (1993 – 1998)

Additional Training

Artificial Intelligence and Data Science for Leaders | The University of Chicago (May 2026).

Management and leadership (Becoming a Leader) program | The University of Chicago (March 2026).

The Longitudinal Data Analysis Using R | Statistical Horizons (Oct 2020).

The Human and Mammalian Genetics and Genomics | The 59th McKusick Short Course, the Jackson Laboratory, Bar Harbor, Maine. USA (July 2018).

Introduction to Java programming | Centennial College, Toronto, ON (Summer 2010).

Research Experience

Lead scientist | IPPH | The University of Chicago

MAY 2026 - Present - Lead interdisciplinary research programs at the interface of human genetics, bioinformatics, statistical genomics, and multi-omics analyses to advance the understanding of complex diseases and therapeutic responses. - Develop and execute innovative research strategies integrating GWAS,

WGS and WES, polygenic risk prediction, fine-mapping, causal inference, and multi-omics analyses to identify genetic and molecular mechanisms underlying disease susceptibility, progression, and treatment outcomes. - Design and oversee scalable computational workflows for the analysis and integration of genomic, transcriptomic, epigenomic, proteomic, metabolomic, and single-cell datasets using high-performance computing and reproducible bioinformatics pipelines. - Apply advanced statistical modeling, machine learning, and data science approaches to analyze large-scale biomedical datasets, derive clinically meaningful insights, and support precision medicine initiatives. - Lead multidisciplinary teams of scientists, bioinformaticians, statisticians, clinicians, and trainees, providing scientific direction, technical mentorship, and project oversight while fostering collaborative and innovative research environments. - Establish and maintain collaborations with academic institutions, healthcare organizations, and industry partners to translate genomic discoveries into diagnostic, prognostic, and therapeutic applications. - Oversee data quality, reproducibility, regulatory compliance, and implementation of best practices for scientific computing and data management. - Secure competitive research funding through grant development, disseminate findings through high-impact peer-reviewed publications and international conferences, and contribute to the development of novel computational methodologies and analytical frameworks that accelerate genomic discovery and enable the translation of multi-omics research into precision health and personalized medicine.

Postdoctoral Fellow | Khan Lab | Faculty of Medicine | The University of Chicago

FEB 2024 - Present - I have Led research investigating germline genetic influences on CAR T-cell therapy outcomes using a large clinico-genomic dataset of DLBCL patients from the University of Chicago Biobank. - I have overseen Implementation of polygenic risk score (PRS) analyses to evaluate associations between inherited variants, CAR T-cell response, and autoimmune traits using PGS Catalog and LDpred2. - I have established machine and deep learning models to prioritize and functionally annotate non-coding genetic variants relevant to T-cell biology. - I have Integrated AI-driven SuSiE and FINEMAP fine-mapping and CTWAS eQTL approaches to identify candidate causal genes associated with poor CAR T-cell therapeutic outcomes. - I have Cross-referenced these causal variants/genes with single-cell atlas expression profiles to assess lineage-specific expression within CD4⁺ and CD8⁺ T-cell subsets. - I have developed unsupervised Leiden clustering and trajectory inference (PAGA) on single-cell RNA-seq data from over 100K cells, leading to the discovery of a novel rare T-cell subtype implicated in CAR T-cell therapy resistance and autoimmune disease progression.

Ph.D. Researcher | Zhai Lab | Faculty of Medicine | Memorial University of Newfoundland

2018 - 2023 - Utilized advanced bioinformatics pipelines and statistical models for GWAS, WGS, and WES analyses to identify novel genetic factors impacting osteoarthritis progression and total joint replacement outcomes. - Performed large-scale metabolomics analysis to identify the endotypes of primary osteoarthritis patients based on their metabolic profiling. - Identified biomarkers linked to muscle strength decline over 10 years. - Published multiple high-impact peer-reviewed papers and presented findings at national and international conferences including Canadian Orthopedic Association meetings, OARSI, and ASHG meetings.

M.Sc. Researcher | Savas Lab | Faculty of Medicine | Memorial University of Newfoundland

2012 - 2016 - Conducted computational prediction and analysis of CNV profiles in colorectal cancer patients using genome-wide SNP genotyping datasets. - Implemented rigorous statistical models and bioinformatics QC pipelines to improve CNV detection accuracy. -Mapped genes and disrupted pathways associated with predicted CNVs to assess their biological implications in colorectal cancer.

Awards and Scholarships

1. The Top Canadian Researchers Award at The Canadian Orthopedic Association Annual meeting, 8-11 June 2022, Quebec City, Canada.

2. My study entitled “*Endotypes of primary osteoarthritis identified by plasma metabolomics analysis*” has been selected among the Canadian Arthritis Society top 10 research advances of 2021. <https://gazette.mun.ca/research/personalized-treatment/>.
3. The division of Biomedical Sciences (Genetics) research forum award, May 2020.
4. The Canadian Arthritis Society PhD Salary award for three years (2019–2022).
5. The Canadian Connective Tissue Conference (CCTC2019) travel award, May 29-31, 2019, McGill University, Montreal, Québec, Canada.
6. The Howard Hughes Medical Institute scholarship to attend the Human and Mammalian Genetics and Genomics, The 59th McKusick Short Course, the Jackson Laboratory, Bar Harbor, Maine, USA (July 2018).
7. The Terry Fox Research Institute (TFRI) and Beatrice Hunter Cancer Research Institute (BHCRI) travel bursary (July 2015).
8. The Canadian Institutes of Health Research (CIHR) and the Institute Community Support (ICS) Travel Award (Spring 2015).
9. The first inaugural MGSS Graduate Travel Award in Medicine, Faculty of Medicine, Memorial University of Newfoundland (2013-2014).

Technical Expertise

Programming and scripting languages: Bash, Perl, Java, R, Python, SQL, Git, Docker.

Bioinformatics tools and software: sequence alignment & mapping (BWA, Bowtie2, SAMtools), variant calling & manipulation (PLINK, GATK, SAMtools), variant annotation & interpretation (ANNOVAR, SnpEff, SnpSift), polygenic risk scores (PRS) analyses (PRS-CS, LDpred2, Lassosum2), fine-mapping analysis (SusieR, FINEMAP).

Machine learning: Random Forest, XGBoost, scikit-learn, Auto-sklearn, TensorFlow, JAX, Pandas, NumPy.

Statistical genetics and genomics: GWAS, WES, WGS, RNA-seq, ChIP-seq, scRNA-seq, PRS. **Databases & biological resources:** SQL, NCBI (dbSNP, dbGaP, GEO, ClinVar), UCSC Genome Browser, Ensembl, KEGG, OpenGWAS, GWAS Catalog, PRS Catalog.

Workflow management: Nextflow, Snakemake, Docker, Singularity/Apptainer, Google Cloud. **Version**

Control: GitHub, GitLab, Git.

Development & Workflow Management: Conda/Bioconda/Mamba, Jupyter Notebooks/JupyterLab, Rstudio, Eclipse.

Computer infrastructure and platforms: Google Cloud Platform, Digital Research Alliance of Canada.

Publications

1. **Salem Werdyani**, Xiaotong Sun, Yuanyuan Zha, Peter Riedell, Justin Kline, Xin He, Aly A. Khan. Association of Germline Genetic Variants with CAR T-Cell Therapy Outcomes in B-Cell Malignancies (**in preparation**).
2. Hélène Paradis*, **Salem Werdyani***, Guangju Zhai, Robert L. Gendron, Reza Tabrizchi, Margaret McGovern, Michael Jumper, Daniel Brinton, William V. Good. Genetic Variants of the Beta-Adrenergic Receptor Pathways as Both Risk and Protective Factors for Retinopathy of Prematurity. *Am. J. Ophthalmology*. 2024, 263, 179–187 (* shared first authors).
3. Sushma Singh*, Cindy Penney*, Anne Griffin, Geoffrey Woodland, **Salem Werdyani**, Tammy Benteau, Nelly Abdelfatah, Jessica Squires, Beverly King, Jim Houston, Matthew J. Dyer, Nicole M. Roslin, Daniel Vincent, Pascale Marquis, Darren D. O’Rielly, Kathy Hodgkinson, Taylor Burt, Ashley Baker, Susan G. Stanton, TerryLynn Young. Highly variable hearing loss due to POU4F3 (c.37del) is revealed by 2 longitudinal, frequency specific

analyses. Manuscript accepted for publication at the European Journal of Human Genetics (EJHG), (* shared first authors).

4. Justin Pater, Cindy Penney, Darren O’Rielly, Anne Griffin, Mordechai Shohat, Ortal Bare, Curtis R French, Sushma Singh, **Salem Werdyani**, Taylor Burt, Nelly Abdelfatah, Jim Houston, Lance Doucette, Jessica Squires, Nicole Roslin, Daniel Vincent, Pascale Marquis, Geoffrey Woodland, Susan Stanton, Terry-Lynn Young. *Autosomal Dominant Non-Syndromic Hearing Loss Maps to DFNA33 (13q34) and Co-Segregates with Splice Site Variants in ATP11A, A Phospholipid Flippase Gene*. Human Genetics 2021. PMID: 35278131, DOI: 10.1007/s00439-022-02444-x.
5. **Salem Werdyani**, Dawn Aitken, Zhiwei Gao, Ming Liu, Edward W. Randell, Proton Rahman, Graeme Jones, Guangju Zhai. *Metabolomic signatures for the longitudinal reduction of muscle strength over 10 years*. Under review in the BMC Skeletal Muscles Journal. PMID: 35130970, DOI: 10.1186/s13395-022-00286-9.
6. Yajun Yu, **Salem Werdyani**, Megan Carey, Patrick Parfrey, Yildiz E. Yilmaz, Sevtap Savas. *A comprehensive analysis of SNPs and CNVs identifies novel markers associated with disease outcomes in colorectal cancer*. Molecular Oncology 2021 Jul 26. PMID: 34309201, DOI: 10.1002/1878-0261.13067.
7. **Salem Werdyani**, Ming Liu, Hongwei Zhang, Guang Sun, Andrew Furey, Edward W. Randell, Proton Rahman, Guangju Zhai. *Endotypes of primary osteoarthritis identified by plasma metabolomics analysis*. Rheumatology (Oxford), 2020. PMID: 33159799, DOI: 10.1093/rheumatology/keaa693.
8. Lesa M. Dawson*, Kerri N. Smith*, **Salem Werdyani**, Robyn Ndikumana, Cindy Penney, Louisa L. Wiede, Kendra L. Smith, Justin A. Pater, Sheila Drover, Terry-Lynn Young, Darren D. O’Rielly. *A dominant RAD51C pathogenic splicing variant predisposes to breast and ovarian cancer in the Newfoundland population due to founder effect*. Molecular Genetics and Genomic Medicine, 2019 Nov 28. PMID: 31782267, DOI: 10.1002/mgg3.1070. (* shared first authors).
9. **Salem Werdyani**, Yajun Yu, Georgia Skardasi, Jingxiong Xu, Konstantin Shestopaloff, Wei Xu, Elizabeth Dicks, Jane Green, Patrick Parfrey, Yildiz E. Yilmaz, and Sevtap Savas. *Germline INDELs and CNVs in a cohort of colorectal cancer patients: their characteristics, associations with relapse-free survival time, and potential time-varying effects on the risk of relapse*. Cancer Medicine, 2017. PMID: 28544645, DOI: 10.1002/cam4.1074.
10. Lydia Dan, **Salem Werdyani**, Jingxiong Xu, Konstantin Shestopaloff, Angela Hyde, Elizabeth Dicks, Ban Younghusband, Jane Green, Patrick Parfrey, Wei Xu, Sevtap Savas. *No associations of a set of SNPs in the Vascular Endothelial Growth Factor (VEGF) and Matrix Metalloproteinase (MMP) genes with survival of colorectal cancer patients*. Cancer Medicine, 2016. PMID: 27334288, DOI: 10.1002/cam4.796.
11. Sevtap Savas, Jingxiong Xu, **Salem Werdyani**, Konstantin Shestopaloff, Elizabeth Dicks, Jane Green, Patrick Parfrey, Roger Green, Wei Xu. *A survival association study of 102 polymorphisms previously associated with survival outcomes in colorectal cancer*. BioMed Research International, Volume 2015 (2015). PMID: 26064972, DOI: 10.1155/2015/968743.

Presented abstracts related to human genetics research

A. Oral presentations:

1. **Werdyani, S.**, Liu, M., Furey, A., Gao, Z., Rahman, P., Zhai, G. (2022). *A genome-wide association study identified novel genes for poor outcome of total joint replacement surgery in osteoarthritis patients*. Podium presentation, Canadian Orthopaedic Research Society (CORS) Top Canadian Research Paper Session –

Founders' Award Candidates, Canadian Orthopaedic Association Annual Meeting, June 8–11, 2022, Québec City, Canada.

2. **Werdyani, S.**, Liu, M., Furey, A., Gao, Z., Rahman, P., Zhai, G. (2022). *Novel genes for poor outcome of total joint replacement surgery in osteoarthritis patients identified by a genome-wide association study*. Presented at the Canadian Arthritis Research Conference – Annual Research Presentation Days, January 25 & 31, 2022.
3. **Werdyani, S.**, Li, Q., Aitken, D., Liu, M., Dohey, A., O'Rielly, D., Rahman, P., Zhai, G. (2021). *Novel genes associated with osteoarthritis identified by whole-exome sequencing analysis*. Presented at the Virtual Canadian Connective Tissue Conference (CCTC), June–August 2021.
4. **Werdyani, S.**, Aitken, D., Gao, Z., Liu, M., Randell, E.W., Rahman, P., Jones, G., Zhai, G. (2021). *Metabolomic signatures for the longitudinal reduction of muscle strength over 10 years*. Osteoarthritis Research Society International (OARSI 2021) Virtual World Congress, April 29–May 1, 2021.
5. **Werdyani, S.**, Aitken, D., Gao, Z., Liu, M., Randell, E.W., Rahman, P., Jones, G., Zhai, G. (2021). *Metabolomic signatures for the longitudinal reduction of muscle strength over 10 years*. Presented at the Arthritis Society and Canadian Rheumatology Association Virtual Conference, February 5, 2021.
6. **Werdyani, S.**, Liu, M., Xie, Z., Furey, A., Gao, Z., Rahman, P., Zhai, G. (2020). *Genes related to muscle strength, behavioral traits, pain response, and inflammation are associated with poor outcome of total joint replacement therapy in primary osteoarthritis patients*. Podium presentation, Top Canadian Research Paper Session, Canadian Orthopaedic Research Society (CORS) Annual Meeting (Virtual), June 19–21, 2020.
7. **Werdyani, S.**, Liu, M., Sun, G., Furey, A., Randell, E., Rahman, P., Zhai, G. (2020). *Plasma metabolomics identified three distinct endotypes of primary osteoarthritis patients*. Accepted for plenary oral presentation, Osteoarthritis Research Society International (OARSI 2020) World Congress on Osteoarthritis, April 30–May 3, 2020, Vienna, Austria. (Conference cancelled due to COVID-19 pandemic.)
8. **Werdyani, S.**, Liu, M., Furey, A., Gao, Z., Rahman, P., Zhai, G. (2019). *Possible genetic predictors for poor outcome of total joint replacement therapy in primary osteoarthritis patients*. Invited keynote presentation, *Seniors Living Well with Arthritis*, Arthritis Society, November 3, 2019, St. John's, Newfoundland, Canada.
9. **Werdyani, S.**, Liu, M., Furey, A., Randell, E., Rahman, P., Zhai, G. (2019). *Endotypes of primary osteoarthritis identified by a metabolomics approach*. Canadian Connective Tissue Conference (CCTC), May 29–31, 2019, McGill University, Montréal, Québec, Canada.
10. Dan, L., **Werdyani, S.**, Xu, J., Shestopaloff, K., Hyde, A., Green, J., Dicks, E., Green, R., Parfrey, P., Xu, W., Savas, S. (2016). *Associations of polymorphisms in VEGF and MMP family genes with overall survival in colorectal cancer*. Oral presentation by Dr. Sevtap Savas, Target Meeting's 4th World Cancer Online Conference, May 17–19, 2016.
11. Yu, Y., **Werdyani, S.**, Skardasi, G., Xu, J., Shestopaloff, K., Xu, W., Dicks, E., Green, J., Yilmaz, Y., Parfrey, P., Savas, S. (2016). *Structural variants in TGFB3, STEAP2, and FILIP1L genes may associate with disease outcomes in colorectal cancer*. Oral presentation by Mr. Yajun Yu, Target Meeting's 4th World Cancer Online Conference, May 17–19, 2016.
12. Savas, S., Xu, J., **Werdyani, S.**, Shestopaloff, K., Dicks, E., Parfrey, P., Green, R., Xu, W. (2014). *Replication of associations of two polymorphisms with survival times in colorectal cancer*. Invited talk, 2nd International Conference on Predictive, Preventive and Personalized Medicine & Molecular Diagnostics, November 3–5, 2014, Las Vegas, USA.

13. **Werdyani, S.**, Xu, J., Shestopaloff, K., Xu, W., Dicks, E., Green, J., Parfrey, P., Green, R., Savas, S. (2014). *Features of copy number variants in a cohort of colorectal cancer patients*. Invited keynote presentation, 5th UAE National Genetic Diseases Conference, September 13–17, 2014, Dubai, United Arab Emirates.
14. **Werdyani, S.**, Xu, J., Shestopaloff, K., Xu, W., Dicks, E., Parfrey, P., Green, R., Savas, S. (2014). *Genes and pathways affected by germline copy number variations in colorectal cancer patients*. Canadian Human and Statistical Genetics Meeting, May 3–6, 2014, Victoria, British Columbia, Canada.
15. Dan, L., Xu, J., **Werdyani, S.**, Shestopaloff, K., Dicks, E., Parfrey, P., Green, R., Xu, W., Savas, S. (2014). *Genetic polymorphisms in matrix metalloproteinase genes MMP8 and MMP27 are associated with overall survival in colorectal cancer*. Oral presentation by Ms. Lydia Dan, Target Meeting's 3rd World Cancer Online Conference, January 21–24, 2014.

B. Poster presentations:

1. **Werdyani, S.**, Sun, X., Zha, Y., Riedell, P., Kline, J., He, X., Khan, A.A. *Association of Germline Genetic Variants with CAR T-Cell Therapy Outcomes in B-Cell Malignancies*. American Association for Cancer Research (AACR 2025), April 25–30, 2025, Chicago, USA.
2. **Werdyani, S.**, Liu, M., Furey, A., Gao, Z., Rahman, P., Zhai, G. *A genome-wide association study identified novel genes associated with osteoarthritis*. Osteoarthritis Research Society International (OARSI 2022) Virtual World Congress, April 7–10, 2022.
3. **Werdyani, S.**, Liu, M., Furey, A., Gao, Z., Rahman, P., Zhai, G. *Novel genes for poor outcome of the total joint replacement surgery in osteoarthritis patients identified by a genome-wide association study*. Canadian Arthritis Research Conference, February 7–8, 2022.
4. Costello, C., Liu, M., **Werdyani, S.**, Rahman, P., Randell, E.W., Zhai, G. *Genome-wide association study identified MC5R as a candidate gene for refractory knee pain in osteoarthritis patients*. Canadian Arthritis Research Conference, February 7–8, 2022.
5. **Werdyani, S.**, Li, Q., Aitken, D., Liu, M., Dohey, A., O'Rielly, D., Rahman, P., Zhai, G. *Whole exome sequencing analysis identified five novel genes associated with osteoarthritis*. American Society of Human Genetics (ASHG), October 18–22, 2021 (virtual).
6. **Werdyani, S.**, Aitken, D., Gao, Z., Liu, M., Randell, E.W., Rahman, P., Jones, G., Zhai, G. *Metabolomic signatures for the longitudinal reduction of muscle strength over 10 years*. Arthritis Society and Canadian Rheumatology Association Virtual Conference, February 16–17, 2021.
7. **Werdyani, S.**, Liu, M., Xie, Z., Furey, A., Gao, Z., Rahman, P., Zhai, G. *Genes related to muscle strength, behavioral traits, pain response, and inflammation are associated with poor outcome of total joint replacement therapy in osteoarthritis*. Canadian Arthritis Research Conference, February 25–26, 2020, Victoria, BC, Canada.
8. **Werdyani, S.**, Dawson, L.M., Smith, K.N., Hansford, S.E., Ndikumana, R., Young, T.L., O'Rielly, D.D. *A RAD51C mutation causing HBOC is expanded in the Newfoundland population likely due to a founder effect*. American Society of Human Genetics (ASHG), October 16–20, 2018, San Diego, CA, USA.
9. **Werdyani, S.**, Dawson, L.M., Smith, K.N., Hansford, S.E., Ndikumana, R., Young, T.L., O'Rielly, D.D. *A pathogenic splicing founder mutation in RAD51C causing hereditary breast and ovarian cancer in Newfoundland*. The Human and Mammalian Genetics and Genomics – 59th McKusick Short Course, July 2018, Jackson Laboratory, Bar Harbor, ME, USA.

10. Dawson, L.M., **Werdyani, S.**, Smith, K.N., Hansford, S.E., Ndikumana, R., Young, T.L., O’Rielly, D.D. *A likely pathogenic founder mutation in RAD51C causing breast and ovarian cancer in Newfoundland*. BRCA Symposium, May 8–11, 2018, Montreal, QC, Canada (poster presentation by Dr. Lesa Dawson).
11. Yu, Y., **Werdyani, S.**, Skardasi, G., Xu, J., Shestopaloff, K., Xu, W., Dicks, E., Green, J., Parfrey, P., Yilmaz, Y., Savas, S. *Copy number variations associated with early relapse in colorectal cancer*. BHCRI/TFRI Cancer Research Conference in Atlantic Canada, November 7–8, 2016, Halifax, NS, Canada (poster presentation by Y. Yu).
12. Yu, Y., **Werdyani, S.**, Skardasi, G., Xu, J., Shestopaloff, K., Xu, W., Dicks, E., Green, J., Yilmaz, Y., Parfrey, P., Savas, S. *Common copy number variations and disease-free survival in colorectal cancer*. Canadian Human and Statistical Genetics Meeting, April 16–19, 2016, Halifax, NS, Canada (poster presentation by Y. Yu).
13. **Werdyani, S.**, Skardasi, G., Xu, J., Shestopaloff, K., Xu, W., Dicks, E., Green, J., Parfrey, P., Green, R., Savas, S. *Copy number variants and colorectal cancer*. Canadian Cancer Research Conference, November 8–10, 2015, Montreal, QC, Canada.
14. **Werdyani, S.**, Skardasi, G., Xu, J., Shestopaloff, K., Xu, W., Dicks, E., Green, J., Parfrey, P., Green, R., Savas, S. *Copy number variants and small insertions/deletions in colorectal cancer patients from Newfoundland*. Canadian Human and Statistical Genetics Meeting, April 18–21, 2015, Vancouver, BC, Canada.
15. Savas, S., Dan, L.A., Xu, J., **Werdyani, S.**, Shestopaloff, K., Dicks, B., Parfrey, P., Green, R., Xu, W. *Genetic polymorphisms and outcome research in cancer: examples from angiogenesis and metastasis genes in colorectal cancer*. International Conference on Predictive, Preventive and Personalized Medicine & Molecular Diagnostics, November 3–5, 2014, Las Vegas, USA (poster presentation by Dr. Sevtap Savas).
16. Dan, L., Xu, J., **Werdyani, S.**, Shestopaloff, K., Dicks, E., Parfrey, P., Green, R., Xu, W., Savas, S. *Prognostic association of polymorphisms in MMP8 and MMP27 genes in colorectal cancer patients*. Canadian Human and Statistical Genetics Meeting, May 3–6, 2014, Victoria, BC, Canada (poster presentation by L. Dan).
17. Dan, L., Xu, J., **Werdyani, S.**, Shestopaloff, K., Dicks, E., Parfrey, P., Green, R., Xu, W., Savas, S. *Genetic polymorphisms in angiogenesis, lymph-angiogenesis, and metastasis pathway genes and disease outcome in colorectal cancer*. Canadian Cancer Research Conference, November 2–6, 2013, Ontario, Canada (poster presentation by L. Dan).

Teaching & Mentorship

- Delivered lectures and developed curriculum for undergraduate, graduate, and medical courses on human molecular genetics, population genetics, genomics, and genetic epidemiology.
- Led and supervised multidisciplinary research teams, guided studies on gene-environment interactions in complex diseases.
- Mentored and trained undergraduate, graduate students and clinical residents, in research design, data analysis, and career development.
- Authored and co-authored more than 10 peer-reviewed publications in high-impact journals, contributing to advancements in genomics and precision medicine.

Biology related work experience

Research associate | Young Lab | Faculty of Medicine | Memorial University of Newfoundland
 Jan 2016 – Sept 2018

Research associate | Savas Lab | Faculty of Medicine | Memorial University of Newfoundland
Dec 2014 – Jan 2016

Research assistance | Ozcelik Lab | Cancer Research Institute | University of Toronto
May 2011 – Aug 2012

Environmental Researcher and Awareness Educator | Environment Agency- Abu Dhabi | UAE
Sept 2000 – Aug 2006

Medical Laboratory Technologist | The Central Diagnostics Laboratory | UAQ Hospital | UAE
Sept 1998 – Dec 2000

Professional Memberships

- The American Society of Human Genetics (ASHG).
- The Osteoarthritis Research Society International (OARSI).
- The American Association for the Advancement of Science (AAAS).
- The Canadian Arthritis Society.
- The Canadian Connective Tissue Society.
- The Beatrice Hunter Cancer Research Institute (BHCI).
- The European Association for Cancer Research.

Certifications

- TCPS2: Tri- Council Policy Statement: Ethical Conduct for Research Involving Human.
- Personal Health Information Act (PHIA) Certification.

Volunteer work

- Team leader and volunteer at the Emirates Red Crescent Authority, Abu Dhabi (2000-2006).
- Team leader of Personal Disaster Assistance with the Canadian Red Cross to improve the lives of vulnerable people by mobilizing the power of humanity (2010, Present).
- Board of directors' member at the Association for Canadian Education Resources (ACER), (2006- Present).
- Participated in the Schoolyard Tree Measuring Project at the Humber College Arboretum (2008-2012).
- Member of Voluntary network of Toronto and Region Conservation Authority (TRCA) (2010-2012).
- Team leader with the Friends of Claireville in co-operation with the TRCA at Claireville conservation area, rivers, and streams (2010-2012).